



PHASE 1 — FOUNDATION

The Kit Comes Out

Inventory the Triggerfish, then make its first connections.

Say Watt Robotics · ROV Team | 2026-2027 MATE ROV Season

CLASS 1.3 · ONE SESSION · MADISON, ALABAMA

THE MEETING

Three hours, the kit comes alive.

0:00	Welcome & Safety Brief	15 min
0:15	Unbox & Inventory the Kit	35 min
0:50	Read the Build Sequence	20 min
1:10	First Real Connections	50 min
2:00	<i>Break</i>	10 min
2:10	Open the Three Records	25 min
2:35	Design the Capsule	15 min
2:50	Wrap-up & homework	10 min

SECTION 01

Welcome & Safety Brief

0:00 - 0:15 · 15 MIN

01

Every meeting we ever hold will begin in exactly this place.

Every meeting opens with a safety brief.

Why we do this

Real engineering and dive teams brief safety before every job — not because something went wrong, but so it never does. We adopt the same habit on day one. Sixty seconds, every meeting, no exceptions.

Each brief answers the same four questions, so the team always knows what to expect.

- 1 What are we doing today?
- 2 What could hurt someone?
- 3 How do we prevent it?
- 4 Who do we tell if something happens?

TODAY'S BRIEF · CLASS 1.3

THE WORK

Unboxing the kit, the first kit electronics, and opening the team's records.

THE HAZARDS

Hot irons again; small parts and sharp tools; lead in solder.

PREVENTION

Irons in stands, fume fan on, glasses down. Count and bag small parts so none walk off. Wash hands after solder.

IF SOMETHING HAPPENS

Cool a burn under running water, tell the advisor, and log it. A missing part gets flagged the same day.

LOOKING AHEAD

Class 1.4 — the multimeter and learning to read a circuit.



SECTION 02

The Kit Comes Out

0:15 - 0:50 · 35 MIN

The Triggerfish leaves the box, gets counted against the manual, and the first real electronics go in.

02

THE KIT COMES OUT

Unbox it. Count every part.

The SeaMATE Triggerfish is our trainer ROV — the kit every kid learns on before we build the competition vehicle. The moment it opens, the Mechanical Lead runs inventory against the manual’s parts list.

One rule: nothing gets built until every part is counted and checked off. A missing part found today is an email; found mid-build, it is a lost week.

QTY	PART	NOTE	✓
3	Thruster motors	Handle by the body, not the leads	☐
1 set	Frame & hardware	Count every screw	☐

DON'T FORGET

It ships without a power source.

The Triggerfish kit does not include a battery. Our 12 V SLA battery, charger, and Powerpole leads are a separate order — and they must be in hand before the wiring lesson.

OWNER Mechanical Lead

STATUS Battery on order · due before wiring



SECTION 03

First Electronics

0:50 - 1:10 · 70 MIN

Irons hot again — the joints we practiced become real connections on the robot.

03

Read the manual. Then wire stage one.

- 1 Read the build order**

Walk the manual's sequence start to finish before touching a part, so everyone knows the whole path.
- 2 Lay out stage one**

Group the parts for the first stage and match each one to the diagram. Ask questions now, not mid-solder.
- 3 Make the first kit joints**

The irons are already hot. The joints we practiced become real connections on the robot.
- 4 Photograph and log it**

Shoot the stage, write what we did and any decision we made. The notebook fills as the robot grows.

Two things to remember. We stop wherever the clock runs out. A clean stopping point always beats a rushed connection.



SECTION 04

The Records Begin Today

2:10 - 2:35 · 25 MIN

Five records that win in April start quietly today — repo, wiki, notebook, photos, outreach.

04

Five records open today and never close.

The build is only half the season. The documentation, the code, and the outreach are scored too — and like a joint, they are honest only if made as we go. Five records start now.

1

GitHub org

The Software Lead creates the team's GitHub organization on student accounts. All code lives here.

2

Project wiki

A shared wiki for how-tos, wiring notes, and decisions the whole team can read and edit.

3

Project notebook

Every decision, with its reason. Written as we go — a notebook from memory is fiction.

4

Photo log

Photograph the work every meeting. These pictures become the marketing display and the report.

5

Outreach log

Log everyone we teach or reach. Outreach is scored and cannot be faked in April.

SECTION 05

The Waterproof Capsule

2:35 - 2:50 · 15 MIN

A five-dollar dress rehearsal for the float that must survive 2.5 meters.

05

Keep a paper towel dry. Underwater.

Each kid designs a small waterproof capsule — a sealed PVC body with a single cable passing through the wall. The test could not be simpler: a paper towel sealed inside has to come out bone dry after a dunk.

It is the same problem the Phase 2 float must solve at 2.5 meters, where a leak ends the mission. Here it costs five dollars and a do-over. Sketch your design today; we cut and seal next week.

One body, one cable penetration, zero leaks. Draw your plan before you leave today.

THE BRIEF

Seal the body. Seal the cable. Prove it stays dry.

Due today: your sketch

Every student designs one. We cut and seal them for real in a later lesson.

SECTION 06

Wrap-Up & Homework

2:50 - 3:00 · 10 MIN

Close the loop: what each student carries out, and what the team carries forward into the water.

06

WRAP-UP & HOMEWORK

Before we meet again.

EVERY STUDENT

- 1 Finish your capsule sketch — one body, one cable, a plan to stay dry.
- 2 Software Lead: confirm everyone can log in to the GitHub org.
- 3 Watch the Class 1.4 pre-watch (Multimeter Basics) when assigned.

THE PARALLEL TRACK · LEADS & ADVISORS

Battery, charger, and leads in hand before the electronics workshop. SeaMATE Simple Circuits Lab Kits on order. Pool prerequisites moving — lifeguard, swim checks, host rules in writing.

NEXT LESSON · CLASS 1.4

Electronics Workshop

The multimeter, switches, simple circuits, and the Electrical Symbols handout — learning to read a circuit.



THE KIT IS ALIVE

It's a robot now.

The Triggerfish is inventoried, its first connections are made, the records are open, and every capsule is designed.

Next, we learn to read it.